

```

public static void pattern9(int n){
    int mid = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i==j && i<=mid || i+j == n-1 && i<=mid ||
            j==mid && i<=mid || i==mid || i-j==mid || i+j == (n-1)+mid){
                System.out.print("* ");
            }else{
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}

```

Pattern 11:

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

```

00 01 02 03 04 05 06 07 08
10 11 12 13 14 15 16 17 18
20 21 22 23 24 25 26 27 28
30 31 32 33 34 35 36 37 38
40 41 42 43 44 45 46 47 48
50 51 52 53 54 55 56 57 58
60 61 62 63 64 65 66 67 68
70 71 72 73 74 75 76 77 78
80 81 82 83 84 85 86 87 88

```

Pattern 10:

```

* * * * *
*           *
*           *
* * * * *
*           *
*           *
* * * * *

```

n=7

```

public static void pattern10(int n){
    int mid = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i==0 && j!=n-1 || i==n-1 && j!=n-1 || j==0 ||
            j==n-1 && i!=0 && i!=mid && i!=n-1 || i==mid && j!=n-1){
                System.out.print("* ");
            }else{
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}

```

```

00 01 02 03 04 05 06
10 11 12 13 14 15 16
20 21 22 23 24 25 26
30 31 32 33 34 35 36
40 41 42 43 44 45 46
50 51 52 53 54 55 56
60 61 62 63 64 65 66

```

Pattern 12:-

```
* * * * *
*           *
*           *
* * * * *
*           *
*           *
*           *
```

n=7

```
public static void pattern12(int n){
    int mid = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i==0 && j != n-1 || j == 0 || i == mid && j != n-1 ||
            i==j && i >= mid || j==n-1 && i< mid && i != 0){
                System.out.print("* ");
            }else{
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Pattern 13:-

```
* * * * *
*           *
*           *
*       * * *
*           *
*           *
*       * * *
*       * * *
```

n=7

```
public static void pattern13(int n){
    int mid = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i == 0 && j!=0 || j ==0 && i!=0 && i!=n-1 || i == n-1 && j!=0 && j!=n-1 ||
            j == n-1 && i > mid && i!=n-1 || i== mid && j >= mid && j!=n-1){
                System.out.print("* ");
            }else{
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Pattern 14:-

```
* * * * *
*           *
*           *
*           *
*           *
*       *
*       *
```

```
public static void pattern14(int n){
    int mid = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i==0 || j==mid && i!=n-1 || i==n-1 && j==mid-1 || i==n-2 && j==mid-2){
                System.out.print("* ");
            }else{
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Pattern 15:-

```
* * * * *
*           *
*           *
*       * * *
*           *
*           *
*       * * *
```

```
public static void pattern15(int n){
    int mid = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i==0 && j!=0 || j==0 && i<mid && i!=0 ||
            i==n-1 && j!=n-1 || j==n-1 && i>mid && i!=n-1
            || i==mid && j!=0 && j!=n-1){
                System.out.print("* ");
            }else{
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
```

Pattern 16:-

```

  * * *
*           *
*           *
*       * * *
  * * * *
                        *
```

```

public static void pattern16(int n){
    int mid = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i==0 && j<n-3 && j!=0 || j==0 && i< n-3 && i!=0 ||
            i==n-3 && j<=n-3 && j!=0 || j==n-3 && i<=n-3 && i!=0 ||
            i==j && i>=mid && j!=n-1){
                System.out.print("* ");
            }else{
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}
```

H/W

```

* * * * * * * * *
* * * * * * * * *
* * * * * * * * *
* * * * * * * * *
* * * * * * * * *
* * * * * * * * *
* * * * * * * * *
* * * * * * * * *
```

Pattern problem using rules.

- Declare the credentials for the first line.
- Create No of row for given input.
- make the credentials (inner loop) with appropriate order.
- update the credentials for the next line.

pattern: 1:-

```

*
* *
* * *
* * * *
* * * * *
```

n = 5

```

public static void pattern1(int n){
    int str = 1; //st1
    for(int i=0;i<n;i++){ //st2
        for(int j=0;j<str;j++){ //st3
            System.out.print("* ");
        }
        str++; //st4
        System.out.println();
    }
}
```

pattern: 2 :-

```
* * * * *
* * * *
* * *
* *
*
```

```
public static void pattern2(int n){
    int str = n; //st1
    for(int i=0; i<n; i++){ //st2
        for(int j=0; j<str; j++){ //st3
            System.out.print("* ");
        }
        str--; //st4
        System.out.println();
    }
}
```

pattern: 3 :-

```
      *
     * *
    * * *
   * * * *
  * * * * *
```

```
public static void pattern3(int n){
    int str = 1, sp = n-1; //st1
    for(int i=0; i<n; i++){ //st2
        for(int j=0; j<sp; j++){
            System.out.print(" ");
        }
        for(int j=0; j<str; j++){
            System.out.print("* ");
        }
        sp--;
        str++;
        System.out.println();
    }
}
```

pattern: 4 :-

```
* * * * *
 * * * *
  * * *
   * *
    *
```

```
public static void pattern4(int n){
    int str = n, sp = 0; //st1
    for(int i=0; i<n; i++){ //st2
        for(int j=0; j<sp; j++){ //st3
            System.out.print(" ");
        }
        for(int j=0; j<str; j++){
            System.out.print("* ");
        }
        sp++; //st4
        str--;
        System.out.println();
    }
}
```

pattern 5:

```
*           *
* *         * *
* * *       * * *
* * * *     * * * *
* * * * *   * * * * *
```

$$n = 5$$

$$sp = 7$$

$$= 10 - 3$$

```
*           *
* *         * *
* * *       * * *
* * * *     * * * *
* * * * *   * * * *
* * * * *   * * * *
* * * * *   * * * * *
```

$$n = 7$$

$$sp = 11 \Rightarrow (n \times 2) - 3$$

$$= 14 - 3$$

```

public static void pattern5(int n){
    int str = 1, sp = (n*2)-3;
    for(int i=0;i<n;i++){
        for(int j=0;j<str;j++){
            System.out.print("* ");
        }
        for(int j=0;j<sp;j++){
            System.out.print("  ");
        }
        if(i==n-1){
            str--;
        }
        for(int j=0;j<str;j++){
            System.out.print("* ");
        }
        System.out.println();
        str++;
        sp-=2;
    }
}

```

Pattern 6 :-

```

      *
    * * *
  * * * * *
* * * * * * *
  * * * * *
    * * * *
      *

```

$n=9$.

```

public static void pattern6(int n){
    int str = 1, sp = n/2;//st1
    for(int i=0;i<n;i++){
        for(int j=0;j<sp;j++){
            System.out.print("  ");
        }
        for(int j=0;j<str;j++){
            System.out.print("* ");
        }
        System.out.println();
        if(i<n/2){
            str+=2;
            sp--;
        }else{
            str-=2;
            sp++;
        }
    }
}

```

Pattern 7 :

```

*   *   *   *   *   *   *
  *   *   *   *   *
    *   *   *
      *
    *   *   *
  *   *   *   *   *
*   *   *   *   *   *

```

$$sp = 0, \quad st\lambda = n$$

$stn = 2$
 $sp++ \rightarrow$ upper part.

str += 2
sp -- → lower part

```
public static void pattern7(int n){
    int str = n, sp = 0; //st1
    for(int i=0;i<n;i++){
        for(int j=0;j<sp;j++){
            System.out.print(" ");
        }
        for(int j=0;j<str;j++){
            System.out.print("* ");
        }
        System.out.println();
        if(i<n/2){
            str-=2;
            sp++;
        }else{
            str+=2;
            sp--;
        }
    }
}
```

Pattern 7 :-

$$sp = n/2 \quad \text{str} = 1$$

```
public static void pattern8(int n){
    int str = 1, sp = n/2;
    for(int i=0;i<n;i++){
        for(int j=0;j<sp;j++){
            if(i==n/2){
                System.out.print("* ");
            }else{
                System.out.print("  ");
            }
        }
        for(int j=0;j<str;j++){
            System.out.print("* ");
        }
        System.out.println();
        if(i<n/2){
            str++;
        }else{
            str--;
        }
    }
}
```

HW:

```

* * * * * * * * * * *
* * * * *      * * * * *
* * * *      * * * * *
* * * *      * * * * *
* * * *      * * * * *
* * *      * * * * *
* *      * * * * *
*      * * * * *
*      * * *

```